Faithful Diffusion-Based Image Super-Resolution

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1 Research problem

Conditional diffusion models give the best perceptual quality known for single-image super-resolution [Saharia et al., 2023], sampling from a learned approximation of the posterior $p(x \mid y)$ over high-resolution images consistent with a low-resolution observation y. At $4\times$ that posterior is broad, so the sampler commits to detail the measurement does not determine: a plausible texture, a sharpened edge, a character that looks legible on no evidence. Perceptually this is an improvement; as a measurement it is fabrication.

Partial forward diffusion [Meng et al., 2022] makes the amount controllable through one scalar $s \in [0, 1]$: noise the bicubic upsample of y to timestep $t_0 = \lfloor sT \rfloor$, then denoise to zero; $s = 0$ returns bicubic, $s = 1$ is full generative sampling. For super-resolution this knob is essentially unmeasured.

We ask: **(Q1)** for a fixed model and fixed y, as s rises, how do perceptual quality, distortion, measurement fidelity and hallucination each move, and does the pattern differ for stochastic texture versus regular man-made structure? **(Q2)** does an explicit reconstruction constraint—projecting each sampler iterate back onto the images consistent with y—*translate* the trade-off curve toward faithfulness, or only move the operating point along it? Blau and Michaeli [2018] prove a trade-off must exist, not that a given sampler sits on its frontier; Q2 asks whether this one does.

2 Why this problem is important

Because the trade-off is unavoidable [Blau and Michaeli, 2018] there is no best operating point, only a choice—made implicitly today, by picking a model and a sampler, with no map of what the alternatives cost. The cost of an invented detail is also asymmetric: a blurry reconstruction is honest about its ignorance, a sharp wrong one is not, and in medical, forensic and scientific imaging the second failure is the damaging one [Tivnan et al., 2024]. Hallucination has meanwhile acquired a precise definition: Cohen et al. [2024] identify it with posterior uncertainty, measurable as disagreement among samples sharing y, which is what makes a systematic sweep possible now.

3 Existing work and its limitations

SR3 [Saharia et al., 2023] established conditional diffusion super-resolution by concatenating the bicubic upsample of y to the denoiser input; it notes that distortion metrics worsen, but samples at one fixed strength. SDEdit [Meng et al., 2022] introduced the strength knob as a realism–faithfulness dial, but for stroke-based editing, with no ground truth and no measurement operator, so it quantifies neither distortion nor hallucination. Consistency-enforcing samplers—DDNM [Wang et al., 2023], diffusion posterior sampling [Chung et al., 2023], ILVR [Choi et al., 2021]—show fidelity to y can be imposed, but each evaluates a single configuration; none asks how the constraint reshapes the trade-off. Closest to our question, Tivnan et al. [2024] note in passing that more modest denoising strengths reduce hallucination, on one medical modality with a single scalar index, no sweep and no intervention tested; and Cohen et al. [2024] supply the uncertainty-based estimator, validated on BSD100 at $4\times$ bicubic, exactly our protocol, but to compare models rather than to trace one across strengths.

The field thus has a theorem, isolated point measurements, a knob borrowed from editing, and one anecdote; missing is a *measured curve* of faithfulness and hallucination against perceptual quality

as continuous functions of strength within one trained model, plus a controlled test of whether data consistency moves it.

4 Proposed solution

Model and knob. One conditional diffusion model in the SR3 formulation [Saharia et al., 2023]: ϵ-prediction DDPM objective [Ho et al., 2020], bicubic upsample of y concatenated channel-wise to the noisy target, trained at $4\times$ on the 800 DIV2K training images [Agustsson and Timofte, 2017] with our own degradation operator. Exactly one model is trained; all baselines are deterministic or pretrained. Sampling is DDIM [Song et al., 2021] with $\eta = 0$ from $t_0 = \lfloor sT \rfloor$, step spacing fixed at $T/50$, so smaller s means strictly fewer network evaluations and compute becomes a reported third axis.

Measurement, per $(s, \text{seed}, \text{image})$: LPIPS [Zhang et al., 2018]; Y-channel PSNR and SSIM [Wang et al., 2004] with a 4-pixel border crop, per the EDSR protocol [Lim et al., 2017]; LR consistency, the PSNR between the bicubic decimation of the output and y; the per-pixel standard deviation across seeds, image-averaged, as hallucination-as-posterior-uncertainty [Cohen et al., 2024]; and median wall-clock. The last two need no ground truth.

Design. $s \in \{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ crossed with five seeds on BSD100 [Martin et al., 2001] and Urban100 [Huang et al., 2015], a contrast pair: invented stochastic texture is perceptually cheap, invented regular structure is not. Set5 and Set14 only confirm the pipeline reproduces published numbers; baselines are bicubic ($s = 0$) and pretrained EDSR [Lim et al., 2017]. We then insert DDNM back-projection [Wang et al., 2023], $\hat{x}_0 \leftarrow \hat{x}_0 - H^{\dagger}(H\hat{x}_0 - y)$, at every step and repeat the identical sweep, so the constraint is the only difference between the curves. Analysis is paired per image: bootstrap intervals, Wilcoxon signed-rank between adjacent strengths with Holm correction [Holm, 1979], Spearman for monotonicity. Hypotheses: **(H1)** LPIPS improves while PSNR and LR consistency degrade monotonically in s; **(H2)** cross-seed deviation rises with s and correlates with the LPIPS gain; **(H3)** back-projection translates the curve, giving equal LPIPS at higher LR consistency.

5 Novelty and significance

The novelty is the measurement, not a new sampler. We are not aware of work that traces the perception–faithfulness trade-off as a continuous function of generative strength within one diffusion model, with hallucination measured as posterior uncertainty rather than inferred from falling PSNR, and that then tests whether a data-consistency constraint translates the curve. We do not claim H1's phenomenon as new—Tivnan et al. [2024] noted it—only its systematic measurement, with uncertainty quantified and multiple-comparison control. The product is an operating-point map: for a faithfulness budget in dB of LR consistency, which s to run and what it costs in LPIPS and in seconds. It separates blur from fabrication, two failures PSNR scores identically, and settles whether back-projection is free or the frontier is tight as Blau and Michaeli [2018] predict.

6 Tentative timeline

Week	Dates	Deliverable and gate
1	Oct 21–27	Degradation, six metrics, data. *Done: 21/21 checks, 28.9 dB*
2	Oct 28–Nov 3	SR3 denoiser, DDIM sampler; overfit-ten and $s{=}1$ gates
3	Nov 4–10	Train on DIV2K; checkpoint chosen on validation split
4	Nov 11–17	Strength knob; main sweep ($6\times5\times200$), resumable
5	Nov 18–24	Back-projection variant; repeat sweep; statistics
6	Nov 25–Dec 1	Figures, write-up, buffer

Fallback. If training has not converged by week 3 we substitute a public pretrained checkpoint; if the sweep runs long we drop to three seeds and BSD100 only. Either outcome of H3 is reportable, so nothing is all-or-nothing.

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